CreditFlow

Automated Loan Restructuring System

Comprehensive Installation & Setup Guide

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1. System Architecture Overview

CreditFlow is built as a distributed three-tier banking application designed to automate, evaluate, and monitor loan restructuring processes. The system comprises:

• **React Frontend Client (Port 5173):** A responsive, glassmorphic UI built in React.js for role-based interactions (Super Admin, Remedial Officer, Credit Officer, Sales Officer, CCPU User, Risk & Compliance).
• **Spring Boot Backend Server (Port 8080):** The core business logic controller using Spring Data JPA, JWT Authentication, and RBAC to manage loan restructuring workflow states.
• **Python AI Document Verification Service (Port 5000):** A Flask service that integrates PyPDF and Gemini Multimodal APIs to perform OCR, signature verification, name-matching, and compliance validation.
• **MySQL Database (Port 3306):** Persists cases, core banking loan records, user profiles, dashboard tasks, notifications, and security audit logs.

2. Prerequisites & Software Downloads

Before installing, download and install the following components on your system:

Software Component	Version Required	Download URL / Reference
Node.js & npm	v18.0.0 or higher	https://nodejs.org/
Java SE Development Kit (JDK)	JDK 17	https://www.oracle.com/java/
MySQL Community Server	v8.0.x	https://dev.mysql.com/downloads/
Python Runtime	v3.9 or higher	https://www.python.org/downloads/
Git Version Control	Latest Stable	https://git-scm.com/

Important Note for JDK 17 Configuration:
Ensure that the **JAVA_HOME** environment variable is configured to point to your JDK installation directory, and that **%JAVA_HOME%\bin** (Windows) or **$JAVA_HOME/bin** (Linux/macOS) is added to your system environment **PATH** variable.

3. Step-by-Step Installation

Step A: Database Creation & Seeding

1. Open your MySQL client (Command Line, Workbench, or DBeaver).
2. Confirm MySQL is running on port **3306**.
3. The application is configured to automatically create the database schema on start. However, you can manually create the schema by running:

```
CREATE DATABASE creditflow;
```

Note: Table generation and default record seeding (defined in schema.sql and data.sql) will trigger automatically during Backend Startup. No manual import is needed.

Step B: Spring Boot Backend Startup

1. Navigate to the backend project folder: **creditflow-backend**.
2. Verify the MySQL credentials in **src/main/resources/application.properties**. By default, it connects to local root user with no password. Change if necessary.
3. Launch the server using the Maven wrapper:

```
# Windows cmd/powershell: .\mvnw.cmd spring-boot:run # macOS/Linux terminal: ./mvnw spring-boot:run
```

The backend will run on **http://localhost:8080**. Watch the logs to confirm the database connects and initialization scripts seed successfully.

Step C: Python AI Service Configuration

1. Navigate to the AI folder: **creditflow-ai**.
2. Initialize a local Python virtual environment to manage dependencies:

```
python -m venv venv .\venv\Scripts\activate # On Windows source venv/bin/activate # On macOS/Linux
```

3. Install the required Python packages:

```
pip install -r requirements.txt
```

4. Create a **.env** file inside the **creditflow-ai** folder and set your Google Gemini API Key:

```
GEMINI_API_KEY=AIzaSyYourKeyHere...
```

5. Start the Flask server:

```
python app.py
```

The service will listen on **http://localhost:5000**. It will fall back to local keyword verification if no Gemini API Key is configured.

Step D: React Frontend Client Startup

1. Navigate to the client folder: **creditflow-app**.
2. Install the frontend Node packages:

```
npm install
```

3. Start the Vite React development server:

```
npm run dev
```

The UI will be hosted at **http://localhost:5173**. Open this address in your web browser.

4. Bypassing Authentication & Offline Mock Mode

To facilitate rapid testing, staging, and demo environments, CreditFlow is equipped with an integrated **Demo Login Bypass** and **Offline Mock Mode**:

1. **DEMO LOGIN Button:** On the login page, you can select any role from the dropdown (e.g. Remedial Officer or Super Admin) and click the green **DEMO LOGIN** button. This bypasses credentials validation instantly and logs you in.

2. **Offline Mock Mode:** If your MySQL database or Spring Boot backend server is not running, the frontend API layer will automatically detect the connection error, transition into Mock Mode, and load all system state from memory matching the SQL seed files. You can completely test case creation, document uploads, compliance tracker monitors, and reports without having the backend server active.

5. Verification Checklist

Ensure correct setup by verifying the following endpoints are listening:

Service	Default Address	Expected Response / Check
React Frontend	http://localhost:5173	Vite Dev Server landing page loads.
REST API Backend	http://localhost:8080	Accessing /api/cases returns 401 (Auth check).
AI Scanner	http://localhost:5000	Flask dev server shows running status.

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